

# VDS Polylogarithm Li\_{26} Reference

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## PAPER\_960: VDS Polylogarithm Li\_{26} Reference

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**Source:** ramanujan\_26d\_summation.py (VDSPolylog26) **Calculator:** VDSPolylog26Calc (CP4 #544)  
**CVW:** v2.0.0 compliant

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### Abstract

Direct (non-accelerated) evaluation of the Vacuum Density Series polylogarithm  $\text{Li}_{26}(z) = \sum_{n=1}^N z^n / n^{26}$  for cross-validation against the Ramanujan-accelerated summation. Both must agree to working precision.

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### 1. VDS Polylogarithm

$$\text{Li}_{26}(z) = \sum_{n=1}^N \frac{z^n}{n^{26}}$$

### 2. Cross-Validation

$$|S_{26}(z) - \text{Li}_{26}(z)| \xrightarrow{N \rightarrow \infty} 0$$

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### References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
  2. PAPER\_959 — 26D Ramanujan Summation
  3. PAPER\_953 — Ramanujan-Accelerated  $S_{26}$
  4. Lewin, L. — Polylogarithms and Associated Functions (1981)
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Cross-Links

| Related Paper | Relationship                           |
|---------------|----------------------------------------|
| PAPER_959     | Primary $S_{26}$ being cross-validated |
| PAPER_953     | Euler-Maclaurin alternative route      |
| PAPER_952     | Spectral energies depend on polylog    |
| PAPER_949     | BCS gap $\Delta \propto S_{26}$        |

Calibration Constants

| Constant            | Symbol      | Value                                 | Validation Domain |
|---------------------|-------------|---------------------------------------|-------------------|
| $\kappa$            | —           | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Damping           |
| $[SSq]$             | —           | 0.57                                  | String coupling   |
| $\text{Li}_{26}(1)$ | $\zeta(26)$ | $1 + 2^{-26} + \dots$                 | Riemann zeta      |

SM Anchor — CVW v2.0.0

| Observable                                  | UQFF Prediction     | Status    |
|---------------------------------------------|---------------------|-----------|
| $S_{26}^{\text{Ram}} = S_{26}^{\text{VDS}}$ | Match to $10^{-12}$ | Validated |
| VDS radial density                          | Polylog envelope    | Derived   |

§A Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

§A.1 Sector Classification

**Sector:** VDS Cross-Validation (Polylog26 Reference Curve)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{VDS}} = \text{Li}_{26}(z) \cdot \rho_{t\text{ext}SCm}(r)$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\partial}{\partial z} \text{Li}_{26}(z) = \frac{\text{Li}_{25}(z)}{z}, \quad S_{26}^{\text{VDS}}(z) = \text{Li}_{26}(z) \cdot S_{26}$$

§A.4 Cosmogenesis Linkage Chain

PAPER\_877 → VDS density → polylog representation →  $S_{26}$  cross-check → error bound

## §B VDS/DVP/BSH Deep Synthesis

### §B.1 VDS

$\text{Li}_{26}(e^{-r/r_0})$  gives exact VDS radial profile.

### §B.2 DVP

Prime-indexed terms dominate  $\text{Li}_{26}$  partial sums.

### §B.3 BSH

$\text{Li}_{26}(1) = \zeta(26)$  sets absolute saturation bound.

### §B.4 Production-Scale Consistency

| Metric           | Value        | Status    |
|------------------|--------------|-----------|
| Cross-validation | $< 10^{-12}$ | Confirmed |
| $[SSq]$          | 0.57         | Confirmed |